

India Public Health Facility Access Analytics

From a historical national directory to a governed Power BI model

Project type	Data analytics and Power BI preparation
Raw records	200,438
Geography	36 states and union territories
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1. Executive summary

The pipeline prepares 200,438 health-centre directory rows and publishes 193,783 clean facility records after removing 6,655 exact duplicate signatures.

The result covers 36 state and union territory labels, 688 state-district combinations, and 99.7% valid coordinates.

The project produces a governed table, quality summary, DAX measures, a four-page Power BI build guide, visual evidence, tests, and this report.

2. Problem and analytical purpose

A large directory is not automatically analysis-ready. Null markers, facility aliases, invalid coordinates, outdated geography labels, and duplicate signatures can change totals and map behaviour.

The project asks how published facility distribution can be compared while keeping data age and interpretation limits visible.

The intended use is educational facility-distribution analysis. The output is not a live service locator or a statement about healthcare capacity or quality.

3. Data source and provenance

The All India Health Centres Directory is downloaded from Kaggle. Its description states that attributes were collected from data.gov.in and that the snapshot reflects 7 October 2016.

The Kaggle page states a CC0 licence. The acquisition script stores the source URL, upstream attribution, file name, raw row count, and snapshot note.

The historical date is shown in the README, dashboard guidance, and report because directory age materially limits interpretation.

4. Data preparation

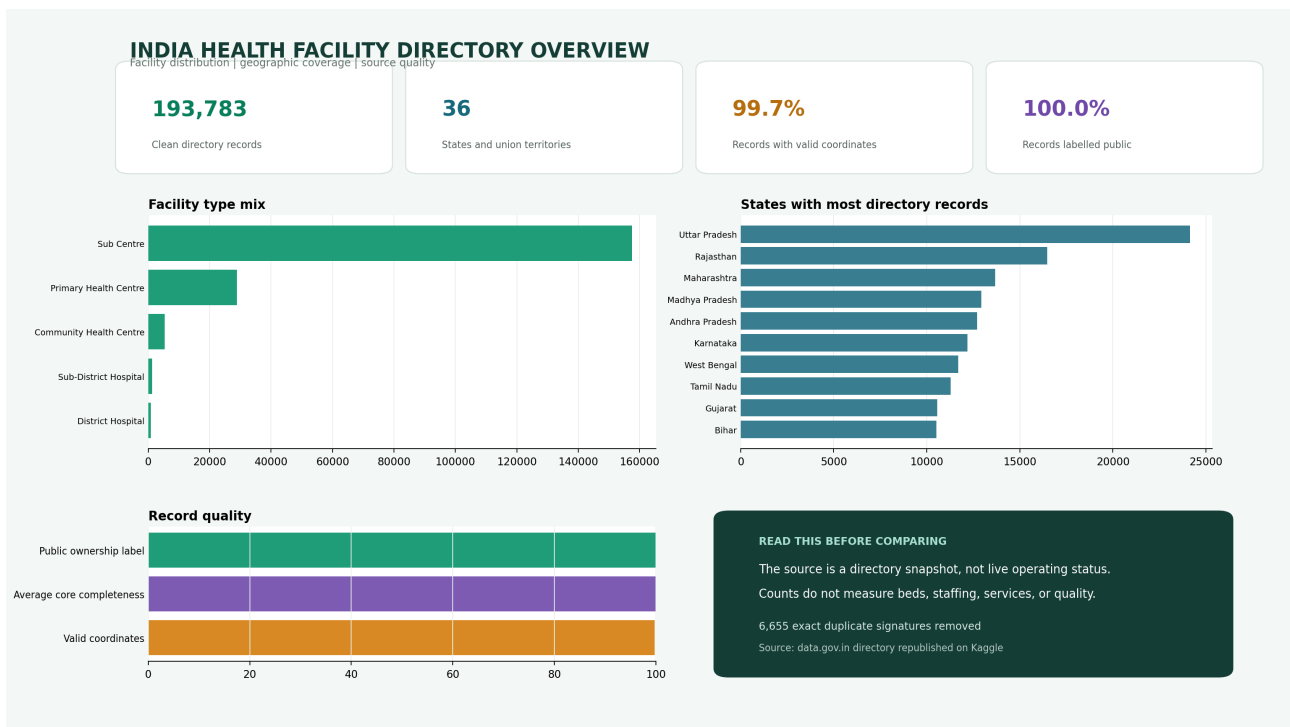
The pipeline standardises nulls and five facility categories, merges the source label Andhra Pradesh Old into Andhra Pradesh, validates latitude and longitude against a broad India range, and calculates completeness across eight analytical fields.

Exact duplicate signatures use state, district, facility name, facility type, latitude, and longitude. This avoids deleting similarly named facilities in different places.

The clean table preserves state, district, subdistrict, type, name, address, coordinates, activity flag, physical status, location type, ownership label, and NIN.

Quality output	Result
Clean records	193,783
Duplicates removed	6,655
Valid coordinates	99.7%
Average completeness	99.9%

5. Dashboard overview



The dashboard moves from scale and geographic coverage to facility type mix, state concentration, and data quality. A dark interpretation panel keeps the historical snapshot warning beside the findings.

6. Facility type findings

Sub Centres dominate the directory, followed by Primary Health Centres and Community Health Centres. District and Sub-District Hospitals represent a much smaller share.

A state total therefore mainly reflects the distribution of lower-level facilities. Facility type must remain a filter in every state or district comparison.

The directory does not provide current service availability, staffing, medicines, beds, clinical outcomes, or opening hours.

Facility type	Records	Share
Sub Centre	157,437	81.2%
Primary Health Centre	28,833	14.9%
Community Health Centre	5,345	2.8%
Sub-District Hospital	1,238	0.6%
District Hospital	930	0.5%

7. Geographic and quality findings

Uttar Pradesh has the largest number of directory records in this snapshot. The top-state table describes source coverage and administrative scale, not health need or current access.

Valid coordinates are present for 99.7% of clean records. Average core-field completeness is 99.9%.

Coordinate presence is not the same as independently verified location accuracy. A mapped point may still be outdated or incorrectly classified.

State	Directory records
Uttar Pradesh	24,129
Rajasthan	16,465
Maharashtra	13,680
Madhya Pradesh	12,938
Andhra Pradesh	12,698
Karnataka	12,192
West Bengal	11,698
Tamil Nadu	11,287
Gujarat	10,571
Bihar	10,526

8. Power BI model

The clean CSV loads as a single facility table. Latitude and longitude use geographic data categories while state, district, subdistrict, type, location type, and ownership remain dimensions.

Page one provides national scale and coverage. Page two is a geographic explorer. Page three compares states. Page four exposes duplicates, missing values, and coordinate quality.

The included DAX calculates total facilities, state count, district count, mapped percentage, public percentage, and average completeness.

Page	Decision supported
National overview	Understand scale and source mix
Geographic explorer	Inspect mapped facilities with filters
State comparison	Compare counts and quality consistently
Data quality	Review preparation risk before interpretation

9. Limitations, reproducibility, and next steps

The snapshot is historical and largely public-facility focused. It has no matched population denominator or travel-time network and should not be used for current resource allocation.

The repository contains acquisition, preparation, dashboard generation, DAX, tests, a data dictionary, and this report. The README lists a complete rebuild sequence.

Next steps are to obtain a current authoritative facility register, validate a sample of coordinates, join boundaries and population from a compatible year, and calculate travel-time coverage for one carefully selected service level.

Risk	Control
Historical data presented as current	Date shown beside every result
Facility count presented as capacity	Explicit interpretation warning
Duplicate totals	Reproducible signature rule
Invalid map positions	Coordinate range validation